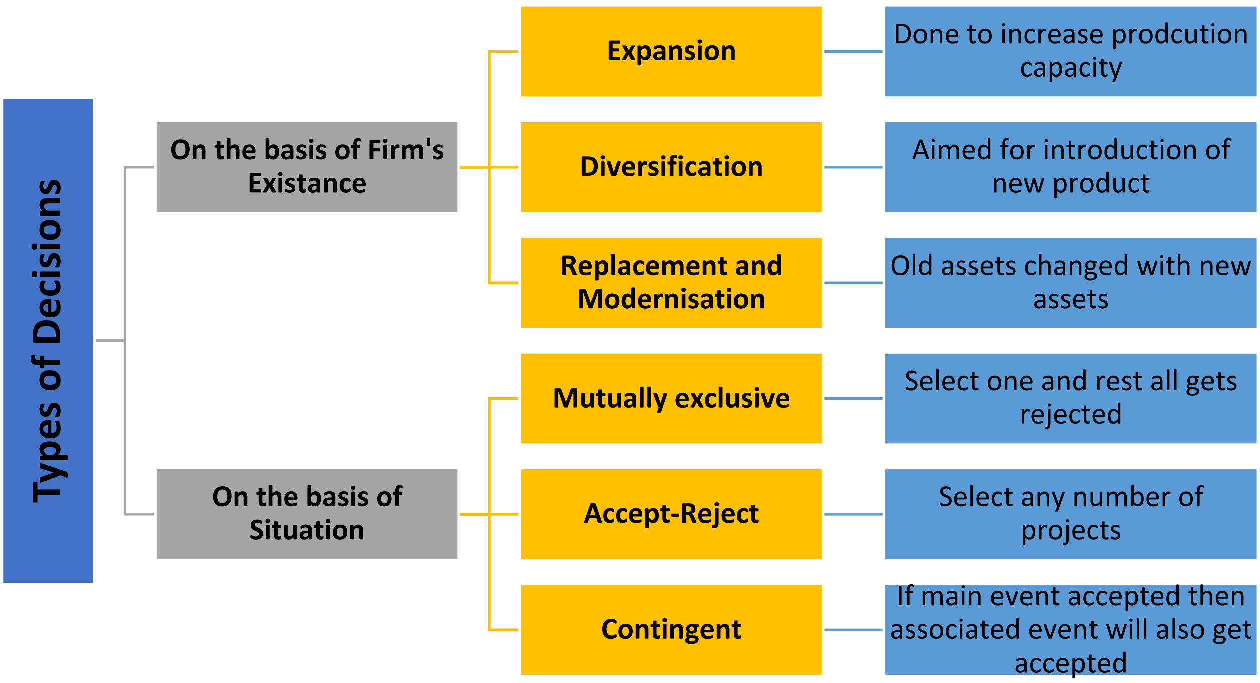
INVESTMENT DECISIONS

1. Capital Budgeting or Investment Decisions

These are related to long term investment decisions or for fixed assets.

2. Types of Decisions



3. Calculation of Book Profit & Cash Flows

Particulars	Amount (₹)
Operating Revenue (e.g. sales etc.)	-
Less: Operating cash costs	-
Profit before depreciation **(or Cash flow before tax)**	-
Less: Depreciation	-
Profit before tax	-
Less: Tax	-
Profit after tax **(Book Profit or Accounting Profit)**	-
Add: Depreciation (Non – Cash Item)	-
Cash Flows after tax	-

4. Calculation of Cash Flows

Cash flows before tax = PBD

Cash flows after tax = PBD – Tax + Tax saving on loss

Cash flows after tax = PAT + Depreciation + Tax saving on loss

Cash flow after tax = Cash flow before tax (1 – t) + (Depreciation × t)

5. Cash flows from sale of assets

Particulars	Amount (₹)
Cost of Assets	-
Less: Accumulated depreciation	-
Book value of assets (A)	-
Sale value of assets (B)	-
Profit/(loss) on sale of assets (A – B)	-
Tax/(tax saving) on sale (C)	-
Net cash flows from sale of assets (D = B – C)	-

6. Concept of Block of Assets

Block is finished

→ No depr. in Year of Sale

→ Capital Gain = Sale Value − Book Value

+ve → Tax Pay

−ve → Tax Saving

→ CF from Sale of Assets = Sale Value − Tax Pay + Tax Saving

Block has balance

→ No Capital Gain to be Computed

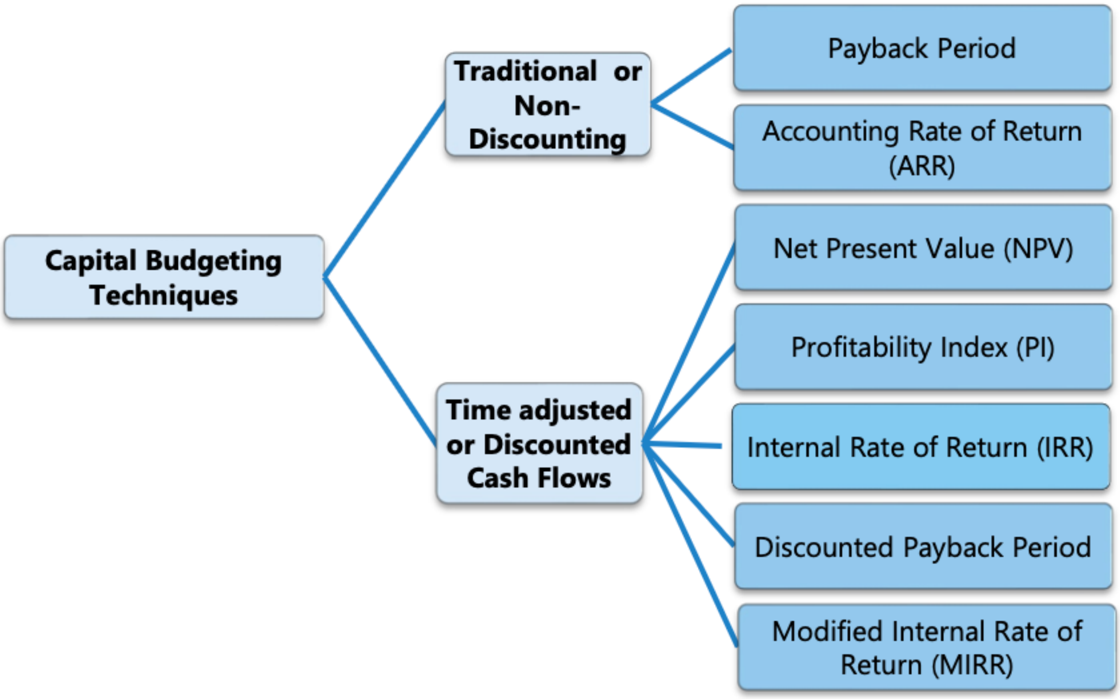
→ Deduct Sale value of assets from Opening block value & Charge depreciation on balance value.

→ Opening Block Value (WDV) = ✓
(+) Purchase of assets (if any) = ✓
(−) Sale of assets = (✓)
✓
(−) Depreciation = ✓
Closing Block value = ✓

(*) Sale value of assets should be net of expenses if any at time of sale.

For solving questions, we have to charge depreciation in year of sale.

7. Techniques of Capital Budgeting



8. Accounting Rate of Return or Average Rate of Return (ARR)

It is the rate of return generated on the funds invested which is based on book profits.

$$ARR = \frac{Average\ PAT}{Average\ Investment} \times 100$$

$$ARR\ on\ Original\ investment = \frac{Average\ PAT}{Original\ Investment} \times 100$$

Where,

Average PAT = $\frac{PAT1+PAT2+\cdots+PATn}{n}$

Average investment of Project = $\frac{AI1+AI2+\cdots+AIn}{n}$

Average investment of a year (AI) = $\frac{Opening+Closing}{2}$

Average Investment of Project = ½(Cost of Project – Scrap value) + Scrap Value + WC

(in case of SLM)

Decision Criteria	Accept	Reject
ARR	Project ARR > Required ARR	Project ARR < Required ARR

9. Payback Period

It is the duration during which cost of project is recovered out of cash inflows.

Payback period (when CFs are equal) $= \frac{\textit{Cost of Project}}{\textit{Annual Cash Inflow}}$

Payback period (when CFs are unequal)

- Calculate cumulative cash flows
- Calculate PBP

Decision Criteria	Accept	Reject
Payback Period	Project PBP < Required PBP	Project PBP > Required PBP

10. Payback Reciprocal or Approx. IRR

It is reverse of payback period.

It is close approximation of IRR.

Payback reciprocal $= \frac{\textit{Average Annual Cash Inflow}}{\textit{Initial Investment}}$

11. Time Value of Money

	Compounding	Discounting
Single Amount	CVF(r, n) = $(1 + r)^n$ Compound Value = Amount × CVF(r, n)	PVF(r, n) = $\frac{1}{(1+r)^n}$ Present Value = Amount × PVF(r, n)
Series of Same amount **or** **Annuity**	CVAF(r, n) = $\frac{(1+r)^n - 1}{r}$ Compound Value = Annual amount × CVAF(r, n)	PVAF = Sum total of PVF Present Value = Annual amount × PVAF(r, n)

12. Discounted Payback Period

It is the duration during which cost of project is recovered from present value of cash inflows of the project.

Step - 1) Calculate discounted cash inflows

Step – 2) Calculate discounted payback period

Decision Criteria	Accept	Reject
Disc. PBP	Project Disc. PBP < Req. PBP	Project Disc. PBP > Req. PBP

13. Net Present Value (NPV)

NPV = PV of cash inflows – PV of cash outflows = PVCI – PVCO

If life of project are different then decision will be based on equivalent annual NPV or equivalent annual PVCO

Equivalent Annual NPV $= \frac{NPV}{PVAF\ for\ Life}$

Equivalent Annual PVCO $= \frac{PVCO}{PVAF\ for\ Life}$

Decision Criteria	Accept	Reject
NPV	NPV > 0 i.e. NPV is +ve	NPV < 0 i.e NPV is -ve

14. Points to Remember (PTRs)

Unless otherwise provided, following points are to be assumed:

(a) Cost of project will be incurred at beginning of the project

(b) Working capital investment will be incurred at beginning of the project

(c) Revenue cash inflows will be at the end of the respective year

(d) 100% of working capital will be realized at end of the project

(e) Sale of assets

a. In case of depreciable assets, sale value = salvage or scrap value

(if scrap value not given than sale value = 0)

b. In case of non-depreciable assets, sale value = cost of assets

(f) Calculate depreciation as per tax laws.

(g) If nothing mentioned then consider books depreciation as tax depreciation.

(h) Losses can be carried forward i.e. tax saving on loss will arise.

15. Treatment of Costs

Relevant
↓
Changes with decision
↓
Consider for decision making
↓
Consider opportunity cost also

Irrelevant
↓
Doesn't change with decision
↓
Ignore for decision making

e.g. → Land already purchased
→ Research exp. already done

16. Treatment of Interest

Don't consider interest for cash inflow calculation as it is already included in cost of capital.

17. Treatment of Subsidy

- Show amount received as negative (-ve) as in PVCO.
- Deduct subsidy from cost of assets while computing depreciation

18. Profitability Index (PI) or Desirability Factor (DF) or Present Value Index or NPV Index Method

It is the amount of cash inflow generated for every rupee of cash outflows.

PI = $\frac{PVCI}{PVCO}$

Decision Criteria	Accept	Reject
PI	PI > 1 i.e. NPV is +ve	PI < 1 i.e NPV is -ve

Question – 1	

ABC Ltd. is considering to purchase a machine which is priced at ₹5,00,000. The estimated life of machine is 5 years and has an expected salvage value of ₹45,000 at the end of 5 years. It is expected to generate revenues of ₹1,50,000 per annum for five years. The annual operating cost of the machine is ₹28,125, Corporate Tax Rate is 20% and the cost of capital is 10%.
You are required to analyse whether it would be profitable for the company to purchase the machine by using;
(i) Payback period Method
(ii) Net Present value method
(iii) Profitability Index Method

Solution

Computation of Annual Cash Flows

Particulars	(₹)
Revenue	1,50,000
Less: Operating Cost	(28,125)
Less: Depreciation $= \frac{(5,00,000 - 45,000)}{5}$	(91,000)
Profit before Tax	30,875
Less: Tax	(6,175)
Profit after Tax	24,700
Add: Depreciation	91,000
Annual Cash Inflows	**1,15,700**

(i) Computation of Payback Period

Year	Cash Flows	Cumulative Present Value
1	1,15,700	1,15,700
2	1,15,700	2,31,400
3	1,15,700	3,47,100
4	1,15,700	4,62,800
5 (Including Salvage)	1,60,700	6,23,500

Amount to be recovered in 5th year cash flow = ₹5,00,000 – ₹4,62,800 = ₹37,200

Payback period = 4 years + $\frac{37,200}{1,60,700}$ = 4.23 years

Since the payback period is less than the life of machinery, the company may purchase the machine.

(ii) Computation of Net Present Value

Year	Cash Flows	PVF @10%	Present Value
0	(5,00,000)	1.000	(5,00,000)
1–5	1,15,700	3.791	4,38,594
5	45,000	0.621	27,941
Net Present Value			(33,465)

Since the net present value (NPV) is negative, the company should not purchase the machine.

(iii) Computation of Profitability Index (PI)

Profitability Index (PI) = $\frac{Sum\ of\ present\ value\ of\ net\ cash\ inflow}{Initial\ cash\ outflow} = \frac{4,38,594+27,941}{5,00,000}$ = 0.93

Since the profitability index is less than 1, the company should not purchase the machine.

Question – 2

Stand Ltd. is contemplating replacement of one of its machines which has become outdated and inefficient. Its financial manager has prepared a report outlining two possible replacement machines. The details of each machine are as follows:

	Machine 1	**Machine 2**
Initial investment	₹ 12,00,000	₹ 16,00,000
Estimated useful life	3 years	5 years
Residual value	₹ 1,20,000	₹ 1,00,000
Contribution per annum	₹ 11,60,000	₹ 12,00,000
Fixed maintenance costs per annum	₹ 40,000	₹ 80,000
Other fixed operating cost per annum	₹ 7,20,000	₹ 6,10,000

The maintenance costs are payable annually in advance. All other cash flows apart from the initial investment assumed to occur at the end of each year. Depreciation has been calculated by straight line method and has been included in other fixed operating costs. The expected cost of capital for this project is assumed at 12%p.a.

Required to compute which machine is more beneficial, using annualized equivalent approach. Ignore tax.

Year	1	2	3	4	5	6
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636	0.567	0.507
$PVIFA_{0.12,t}$	0.893	1.690	2.402	3.038	3.605	4.112

Solution

Statement of Calculation of Cash Flows of Machine-1

Particulars	Year 0	Year 1	Year 2	Year 3
Initial investment	(12,00,000)	-	-	-
Contribution	-	11,60,000	11,60,000	11,60,000
Fixed maintenance cost	(40,000)	(40,000)	(40,000)	-
Other fixed operating cost*	-	(3,60,000)	(3,60,000)	(3,60,000)
Residual value	-	-	-	1,20,000
Net Cash flow	(12,40,000)	7,60,000	7,60,000	9,20,000

*Other fixed operating cost (excluding depreciation0 = 7,20,000 - $\left(\frac{12,00,000-1,20,000}{3}\right)$ = 3,60,000

Statement of Calculation of Cash Flows of Machine-2

Particulars	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Initial invest.	(16,00,000)	-	-	-	-	-
Contribution	-	12,00,000	12,00,000	12,00,000	12,00,000	12,00,000
Fixed maint. Cost	(80,000)	(80,000)	(80,000)	(80,000)	(80,000)	(80,000)
Other fixed operating cost*	-	(3,10,000)	(3,10,000)	(3,10,000)	(3,10,000)	(3,10,000)
Residual value	-	-	-	-	-	1,00,000
Net Cash flow	(16,80,000)	8,10,000	8,10,000	8,10,000	8,10,000	9,10,000

*Other fixed operating cost (excluding depreciation0 = 6,10,000 - $\left(\frac{16,00,000-1,00,000}{5}\right)$ = 3,10,000

Statement of NPV

		Machine 1		Machine 2	
Year	**PVF@12%**	**Cash Flow**	**Present Value**	**Cash Flow**	**Present Value**
0	1.000	(12,40,000)	(12,40,000)	(16,80,000)	(16,80,000)
1	0.893	7,60,000	6,78,680	8,10,000	7,23,330
2	0.797	7,60,000	6,05,720	8,10,000	6,45,570
3	0.712	9,20,000	6,55,040	8,10,000	5,76,720
4	0.636	-	-	8,10,000	5,15,160
5	0.567	-	-	9,10,000	5,61,330
	NPV		6,99,440		13,42,110
	PVAF		2.402		3.605
			2,91,191		3,72,291

Machine 2 is better as it has more equivalent annualized NPV.

Calculation of Sensitivity

Difference in equivalent annualized NPV = 3,72,291 – 2,91,191 = ₹ 81,100

Contribution of Machine 1 = ₹ 11,60,000

Sensitivity relating to contribution of machine 1 = $\frac{81,100}{11,60,000} \times 100 = 7\%$

Question – 3	

CK Ltd. is planning to buy a new machine. Details of which are as follows:

Cost of the Machine at the commencement	₹ 2,50,000
Economic Life of the Machine	8 years
Residual Value	Nil
Annual Production Capacity of the machine	1,00,000 units
Estimated Selling Price per unit	₹ 6
Estimated annual fixed cost (excluding depreciation)	₹ 1,00,000
Estimated variable cost per unit (excluding depreciation)	₹ 3
Advertisement expenses in 1st year in addition of annual fixed cost	₹ 20,000

Maintenance expenses in 5th year in addition of annual fixed cost ₹ 30,000
Cost of capital 12%
Ignore tax

Analyse the above mentioned proposal using the Net Present Value Method and advice.

PV Factor at 12% are as under:

Year	1	2	3	4	5	6	7	8
PV Factor	0.893	0.797	0.712	0.636	0.567	0.507	0.452	0.404

Solution

Statement of Present Value of Cash Flows

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8
Units	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000
Contribution per unit (6-3)	3	3	3	3	3	3	3	3
Total Contribution	3,00,000	3,00,000	3,00,000	3,00,000	3,00,000	3,00,000	3,00,000	3,00,000
(-) Fixed Cost	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000
(-) Advert.	20,000	-	-	-	-	-	-	-
(-) Maint.	-	-	-	-	30,000	-	-	-
Profit Before Dep. or CF	1,80,000	2,00,000	2,00,000	2,00,000	1,70,000	2,00,000	2,00,000	2,00,000
PVF @ 12%	0.893	0.797	0.712	0.636	0.567	0.507	0.452	0.404
Present Value	1,60,740	1,59,400	1,42,400	1,27,200	96,390	1,01,400	90,400	80,800

Total Present value of cash inflows = 9,58,730 (from above table)

NPV = PVCI – PVCO = 9,58,730 – 2,50,000 = ₹ 7,08,730

It is recommended to accept the proposal as it has positive NPV.

Question – 4

SK Ltd. is planning to introduce a new product with a project life of 8 years. Initial equipment cost will be ₹3.5 crores. Additional equipment cost ₹25,00,000 will be purchased at the end of the third year from the cash inflow of this year. At the end of 8 years, the original equipment will have no resale value, but additional equipment can be sold for ₹2,50,000. A working capital of ₹40,00,000 will be needed and it will be released at the end of eight year. The project will be financed with sufficient amount of Equity capital. The sales volumes over eight years have been estimated as follows:

Year	1	2	3	4 & 5	6 – 8
Units	72,000	1,08,000	2,60,000	2,70,000	1,80,000

A sales price of ₹240 per unit is expected and variable expenses will amount to 60% of sales revenue. Fixed cash operating costs will amount to ₹36,00,000 per year. The loss of any year will be set off from the profits of subsequent two years. The company is subject to 30% tax rate and consider 12% to be an appropriate after tax cost of capital for this project. The company follows straight line method of depreciation. Required: Calculate the net present value of the project and advise the management to take appropriate decision.

Note: The PV Factors at 12% are:

Year	1	2	3	4	5	6	7	8
PVIF	0.893	0.797	0.712	0.636	0.567	0.507	0.452	0.404

Solution

Statement of PVCO **(Rs. In Lakhs)**

Particulars	Time	Amount	PVF@12%	PV
Initial equipment cost	0	350	1	350
Working capital	0	40	1	40
			PVCO	**390**

Statement of PVCI **(Rs. In lakhs)**

Particulars	Yr. 1	Yr. 2	Yr. 3	Yr. 4 & 5	Yr. 6-8
No. of units	0.72	1.08	2.60	2.70	1.80
Contribution per unit (240 – 60%)	96	96	96	96	96
Total Contribution	69.12	103.68	249.6	259.2	172.8
(-) Fixed cost	(36)	(36)	(36)	(36)	(36)
PBD	33.12	67.68	213.6	223.2	136.8
(-) Depreciation	(43.75)	(43.75)	(43.75)	(43.75)	(43.75)
(-) Additional Dep.	-	-	-	(4.5)	(4.5)
PBT (A)	(10.63)	23.93	169.85	174.95	88.55
(-) Setoff	-	(10.63)	-	-	-
(+) Carry forward	10.63	-	-	-	-
Taxable profit	-	13.30	169.85	174.95	88.55
Tax (B)	-	(3.99)	(50.955)	(52.485)	(26.565)
PAT (A – B)	(10.63)	19.94	118.895	122.465	61.985
(+) Dep	43.75	43.75	43.75	43.75	43.75
(+) Add. Dep.	-	-	-	4.5	4.5
CF from operations	33.12	63.69	162.645	170.715	110.235
(-) Equip. purchase	-	-	(25)	-	-

CFs	33.12	63.69	137.645	170.715	110.235
PVF@12%	0.893	0.797	0.712	1.203	1.363
PV	29.576	50.761	98.003	205.370	150.250

Total PV of CF from operations	=	₹533.961 lakh
(+) PV of WC relaization (40 lakh × 0.404)	=	₹16.160 lakhs
(+) PV of Assets sale (2.5 lakh × 0.404)	=	₹1.010 lakh
Total PVCI	=	₹551.131 lakh
(-) PVCO	=	₹390.000 lakh
NPV	=	₹161.131 lakh

Since NPV is positive, thus recommended to accept the project.

Question – 5

A chemical company is presently paying an outside firm ₹ 1 per gallon to dispose off the waste material resulting from its manufacturing operations. At normal operating capacity, the waste is about 50,000 gallons per year.

After spending ₹ 60,000 on research, the company discovered that the waste could be sold for ₹ 10 per gallon if it was processed further. Additional processing would however, require an investment of ₹ 6,00,000 in new equipment, which would have an estimated life of 10 years with no salvage value. Depreciation would be calculated by straight line method.

Except for the costs incurred in advertising ₹ 20,000 per year, no change in the present selling and administration expenses is expected, if the new product is sold. The details of additional processing costs are as follows:

Variable – ₹ 5 per gallon of waste put into process

Fixed (excluding depreciation) – ₹ 30,000 per year

In costing the new product, general administrative overheads will be allocated at the rate of ₹ 2 per gallon. There will be no losses in processing and it is assumed that the total waste processed in a given year will be sold in that year. Estimates indicate that 40,000 gallons, of the product could be sold each year. The management when confronted with the choice of disposing off the waste or processing it further and selling it, seeks your advice. Which alternative would you recommend? Assume that the firm's cost of capital is 15% and it pays on an average 35% tax on its income.

Note: Present value of annuity of ₹ 1 at 15% rate of discount for 10 years is 5.019.

Solution

Statement of NPV

Particulars	Amount
Sales (40,000 × 10)	4,00,000
(-) Variable cost (40,000 × 5)	(2,00,000)
(-) Fixed cost	(30,000)
(-) Advertisement cost	(20,000)
(+) Saving in disposal cost (50,000 – 10,000)	40,000
Profit before depreciation (A)	1,90,000
(-) Depreciation (6,00,000 ÷ 10)	60,000
Profit before tax	1,30,000
(-) Tax @ 35%	45,500
Profit after tax	84,500
(+) Depreciation	60,000
Cash inflows	1,44,500
$PVAF_{(15\%, 10 \text{ years})}$	5.019
PVCI	7,25,246
Initial Investment – PVCO	6,00,000
NPV	**1,25,246**

Question – 6

Alpha limited is a manufacturer of computers. It wants to introduce artificial intelligence while making computers. The estimated annual saving from introduction of the artificial intelligence (AI) is as follows:

- Reduction of five employees with annual salaries of ₹ 3,00,000 each.
- Reduction of ₹ 3,00,000 in production delays caused by inventory problem
- Reduction in lost sales ₹ 2,50,000 and
- Gain due to timely billing ₹ 2,00,000

The purchase price of the system for installation of artificial intelligence is ₹ 20,00,000 and installation cost is ₹ 1,00,000. 80% of the purchase price will be paid in the year of purchase and remaining will be paid in next year. The estimated life of the system is 5 years and it will be depreciated on a straight-line basis.

However, the operation of the new system requires two computer specialists with annual salaries of ₹ 5,00,000 per person.

In addition to above, annual maintenance and operating cost for five years are as below:

(Amount in ₹)

Year	1	2	3	4	5
Maintenance & Operating cost	2,00,000	1,80,000	1,60,000	1,40,000	1,20,000

Maintenance and operating cost are payable in advance.

The company's tax rate is 30% and its required rate of return is 15%.

Year	1	2	3	4	5
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683	0.621
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636	0.567
$PVIF_{0.15,t}$	0.870	0.756	0.658	0.572	0.497

Evaluate the project by using Net Present Value and Profitability Index.

Solution

Calculation of Cash Flows

Particulars	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Saving in Salaries		15,00,000	15,00,000	15,00,000	15,00,000	15,00,000
Reduction in production delays		3,00,000	3,00,000	3,00,000	3,00,000	3,00,000
Reduction in lost sales		2,50,000	2,50,000	2,50,000	2,50,000	2,50,000
Gain due to Timely Billing		2,00,000	2,00,000	2,00,000	2,00,000	2,00,000
Salary to computer specialist		(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)
Maintenance & Operating cost		(2,00,000)	(1,80,000)	(1,60,000)	(1,40,000)	(1,20,000)
Depreciation		(4,20,000)	(4,20,000)	(4,20,000)	(4,20,000)	(4,20,000)
Profit before tax		6,30,000	6,50,000	6,70,000	6,90,000	7,10,000
Less: Tax @ 30%		(1,89,000)	(1,95,000)	(2,01,000)	(2,07,000)	(2,13,000)
Add: Depreciation		4,20,000	4,20,000	4,20,000	4,20,000	4,20,000
Add: Maintenance & Operating cost		2,00,000	1,80,000	1,60,000	1,40,000	1,20,000
Less: Maintenance & Operating cost	(2,00,000)	(1,80,000)	(1,60,000)	(1,40,000)	(1,20,000)	-
Net CF	(2,00,000)	8,81,000	8,95,000	9,09,000	9,23,000	10,37,000

Statement of NPV

Particulars	Time	PVF	Amount	Present Value
Initial Investment	0	1	16,00,000	16,00,000
Installation expenses	0	1	1,00,000	1,00,000
Instalment of Purchase Price	1	0.870	4,00,000	3,48,000

			PVCO	**20,48,000**
Cash flows	0	1	(2,00,000)	(2,00,000)
	1	0.870	8,81,000	7,66,470
	2	0.756	8,95,000	6,67,620
	3	0.658	9,09,000	5,98,122
	4	0.572	9,23,000	5,27,956
	5	0.497	10,37,000	5,15,389
			PVCI	**28,84,557**
NPV (PVCI – PVCO)				**8,36,557**
Profitability Index (PVCI÷PVCO)				**1.41**

Since, NPV is positive and Profitability index is greater than one, thus it is recommended to introduce the system.

Question – 7	

SK Ltd. is a News broadcasting channel having its broadcasting Centre in Mumbai. There are total 200 employees in the organisation including top management. As a part of employee benefit expenses, the company serves tea or coffee to its employees, which is outsourced from a third party. The company offers tea or coffee three times a day to each of its employees. 120 employees prefer tea all three times, 40 employees prefer coffee all three times and remaining prefer tea only once in a day. The third party charges ₹ 10 for each cup of tea and ₹ 15 for each cup of coffee. The company works for 200 days in a year.

Looking at the substantial amount of expenditure on tea and coffee, the finance department has proposed to the management on installation of a master tea and coffee vending machine which will cost ₹ 10,00,000 with a useful life of five years. Upon purchasing the machine, the company will have to enter into an annual maintenance contract with the vendor, which will require a payment of ₹ 75,000 every year. The machine would require electricity consumption of 500 units p.m. and current incremental cost of electricity for the company is ₹ 12 per unit. Apart from these running costs, the company will have to incur the following consumables expenditure also:

(a) Packets of coffee beans at a cost of ₹ 90 per packet
(b) Packet of tea powder at a cost of ₹ 70 per packet
(c) Sugar at a cost of ₹ 50 per kg
(d) Milk at a cost of ₹ 50 per litre
(e) Paper cup at a cost of 20 paise per cup.

Each packet of coffee beans would produce 200 cups of coffee and same goes for tea powder packet. Each cup of tea or coffee would consist of 10g of sugar on an average and 100ml of milk.

The company anticipates that due to ready availability of tea and coffee through vending machines its employees would end up consuming more tea and coffee.

It estimates that the consumption will increase by on an average 20% for all class of employees. Also, the paper cups consumption will be 10% more than the actual cups swerved due to leakages in them.

The company is in the 25% tax bracket and has a current cost of capital at 12% per annum. Straight line method of depreciation is allowed for the purpose of taxation. You as a financial consultant is required to advise on the feasibility of acquiring the vending machine.
PV factors @12%:

Year	1	2	3	4	5
PVF	0.8929	0.7972	0.7118	0.6355	0.5674

Solution

A. Computation of CFAT (Year 1 to 5)

Particulars		Amount (₹)
(a) Savings in existing Tea & Coffee charges [(120×10×3) + (40×15×3) + (40×10×1) × 200 days]		11,60,000
(b) AMC of machine		(75,000)
(c) Electricity charges	500 × 12 × 12	(72,000)
(d) Coffee Beans	(W.N.) 144 × 90	(12,960)
(e) Tea Powder	(W.N.) 480 × 70	(33,600)
(f) Sugar	(W.N.) 1248 × 50	(62,400)
(g) Milk	(W.N.) 12480 × 50	(6,24,000)
(h) Paper Cup	(W.N.) 1,37,280 × 0.2	(27,456)
(i) Depreciation	1,00,000 □ 5	(2,00,000)
Profit before Tax		**52,584**
(-) Tax @ 25%		(13,146)
Profit after Tax		**39,348**
Depreciation		2,00,000
CFAT		2,39,438

B. Computation of NPV

Year	Particulars	CF	PVF @ 12%	PV
0	Cost of machine	(10,00,00)	1	(10,00,000)
1 – 5	CFAT	2,39,438	3.6048	8,63,126
Net Present Value				(1,36,874)

Since NPV of the machine is negative, it should not be purchased.
Working Note:
Computation of Qty of consumable
No. of Tea Cups = [(120 × 3 × 200 days) + (40 × 1 × 200 days) × 1.2 = 96,000
No. of Coffee cups = 40 × 3 × 200 days × 1.2 = 28,800
No. of coffee beans packet = $\frac{28,800}{200}$ = 144

No. of Tea Powder Packets $= \frac{96{,}000}{200} = 480$

Qty of Sugar $= \frac{(96{,}000+28{,}800)6{,}000}{1{,}000\text{ g}} = 1248\text{kgs}$

Qty of Milk $= \frac{(96{,}000+28{,}800)6{,}000}{1{,}000\text{ml}} = 12{,}480$ litres

No. of paper cups $= (96{,}000 + 28{,}800) \times 1.1 = 1{,}37{,}280$

Question – 8

A hospital is considering to purchase a diagnostic machine costing ₹ 80,000. The projected life of the machine is 8 years and has an expected salvage value of ₹ 6,000 at the end of 8 years. The annual operating cost of the machine is ₹ 7,500. It is expected to generate revenues of ₹ 40,000 per year for 8 years. Presently the hospital is outsourcing the diagnostic work and is earning commission income of ₹ 12,000 per annum. Consider tax rate of 30% and discounting rate as 10%.

Advise, whether it would be profitable for the hospital to purchase the machine?

Give your recommendations as per Net Present Value method and Present Value Index under below mentioned two situations:

(a) If commission income of ₹ 12,000 p.a. is before taxes

(b) If commission income of ₹ 12,000 p.a. is net of taxes

Given:

t	1	2	3	4	5	6	7	8
PVIF(t,10%)	0.909	0.826	0.751	0.683	0.621	0.564	0.513	0.467

Solution

Analysis of Investment Decisions

Determination of Cash inflows	**Situation-(i) Commission Income before taxes**	**Situation-(ii) Commission Income after taxes**
Cash flow up-to 7th year:		
Sales Revenue	40,000	40,000
Less: Operating Cost	(7,500)	(7,500)
	32,500	32,500
Less: Depreciation (80,000 – 6,000) ÷ 8	(9,250)	(9,250)
Net Income	23,250	23,250
Tax @ 30%	(6,975)	(6,975)
Earnings after Tax (EAT)	16,275	16,275
Add: Depreciation	9,250	9,250
Cash inflow after tax per annum	25,525	25,525
Less: Loss of Commission Income	(8,400)	(12,000)

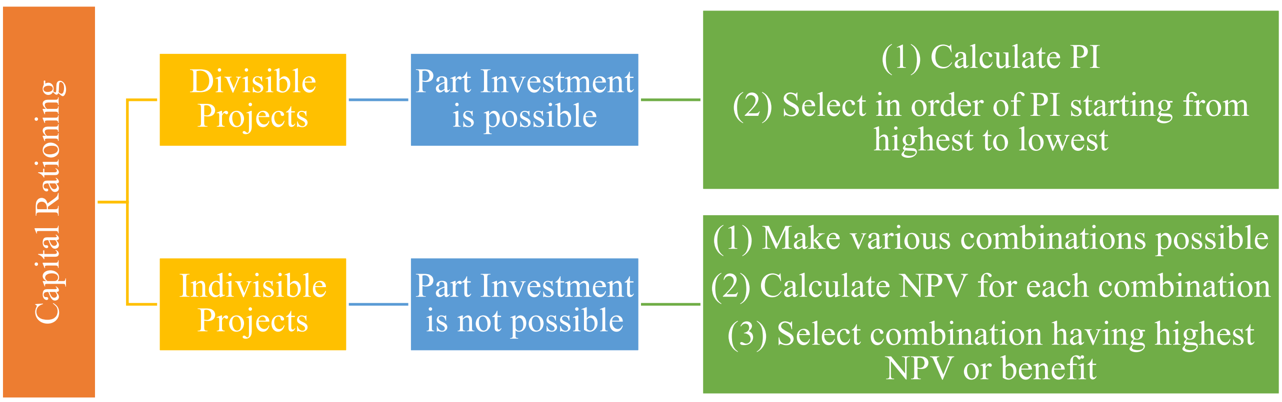
Net Cash inflow after tax per annum	17,125	13,525
In 8th Year		
Net Cash inflow after tax	17,125	13,525
Add: Salvage Value of Machine	6,000	6,000
Net Cash inflow in year 8	**23,125**	**19,525**

Calculation of NPV and Profitability Index

	Particulars	PV factor @10%	Situation-(i) [Commission Income before taxes]	Situation-(ii) [Commission Income after taxes]
A	Present value of cash inflows (1st to 7th year)	4.867	83,347.38 (17,125 × 4,867)	65,826.18 (13,525 × 4.867)
B	Present value of cash inflow at 8th year	0.467	10,799.38 (23,125 × 0.467)	9,118.18 (19,525 × 0.467)
C	PV of cash inflows		94,146.76	74,944.36
D	Less: Cash Outflow	1.00	(80,000)	(80,000)
E	**Net Present Value (NPV)**		**14,146,76**	**(5,055.64)**
F	**PI = (C ÷ D)**		**1.18**	**0.94**

Recommendation: The hospital may consider purchasing of diagnostic machine in situation (i) where commission income is 12,000 before tax as NPV is positive and PI is also greater than 1. Contrary to situation (i), in situation (ii) where the commission income is net of tax, the recommendation is reversed to not purchase the machine as NPV is negative and PI is also less than 1.

19. Capital Rationing



Question – 9

A company has ₹ 1,00,000 available for investment and has identified the following four investments in which to invest.

Project	Investment (₹)	NPV (₹)
C	40,000	20,000
D	1,00,000	35,000
E	50,000	24,000
F	60,000	18,000

You are required to optimize the returns from a package of projects within the capital spending limit if:

(a) The projects are independent of each other and are divisible

(b) The projects are not divisible

Solution

(a) **Computation of NPV per ₹ 1 of investment and Ranking of Projects**

Project	Investment (₹)	NPV (₹)	NPV per ₹ 1 invested (₹)	Ranking
C	40,000	20,000	0.50	1
D	1,00,000	35,000	0.35	3
E	50,000	24,000	0.48	2
F	60,000	18,000	0.30	4

Calculation of Package of Projects

Project	Investment (₹)	NPV (₹)
C	40,000	20,000
E	50,000	24,000
D (1/10th of Project)	10,000	3,500
Total	1,00,000	47,500

The company would be well advised to invest in Project C, E and D (1/10th) and reject Project F to optimize return within the amount of ₹ 1,00,000 available for investment.

(b) **Calculation of Package of Projects**

Package of Project	Investment (₹)	NPV (₹)
C and E	90 000 (40,000 + 50,000)	44,000 (20,000 + 24,000)
C and F	1,00,000 (40,000 + 60,000)	38,000 (20,000 + 18,000)
Only D	1,00,000	35,000

The company would be well advised to invest in Projects C and E to optimize return within the amount of ₹ 1,00,000 available for investment.

20. Internal Rate of Return (IRR)

- Calculate NPV at discount rate given in question (can start with any % given in ques.)
- If NPV is +ve, increase rate to make it -ve.
- If NPV is -ve, decrease rate to make it +ve.
- IRR = $LR + \left[\frac{LR\ NPV}{(LR\ NPV - HR\ NPV)}\right](HR - LR)$

Decision Criteria	Accept	Reject
IRR	IRR > COC	IRR < COC

21. NPV vs IRR

NPV is superior to IRR due to:

(a) Reinvestment rate assumption

- NPV assume it to be at cost of capital
- IRR assume it to be at IRR

(b) Multiple IRR can be computed with same data but it is not possible with NPV

Question – 10

Given below are the data on a capital project 'S':

Annual cost saving	₹ 60,000
Useful life	4 years
Internal rate of return	15%
Profitability index	1.064
Salvage value	0

You are required to calculate for this project S:

(a) Cost of project
(b) Payback period
(c) Cost of capital
(d) Net Present Value

Given the following table of discount factors:

Discounting Factor	15%	14%	13%	12%
1 year	0.869	0.877	0.885	0.893
2 year	0.756	0.769	0.783	0.797
3 year	0.658	0.675	0.693	0.712

4 year	0.572	0.592	0.613	0.636
	2.855	2.913	2.974	3.038

Solution

(a) **Cost of Project S:**

At 15% IRR,

PV of cash inflows = PV of cash outflows

Annual cash inflow × $PVAF_{(15\%,\ 4)}$ = PV of cash outflows

60,000 × 2.855 = PV of cash outflows

PV of cash outflows = ₹ 1,71,300

∴ Cost of project = ₹ 1,71,300

(b) Payback period = $\frac{Cost\ of\ project}{Annual\ cash\ inflow} = \frac{1,71,300}{60,000}$ = 2.855 years

(c) **Cost of capital of Project S:**

Profitability index = $\frac{PV\ of\ cash\ inflows}{PV\ of\ cash\ outflows}$

$1.064 = \frac{Annual\ cash\ inflows \times PVAF}{1,17,300}$

Annual cash inflows × PVAF = 1,82,263.20

60,000 × PVAF = 1,82,263.20

PVAF = $\frac{1,82,263.20}{60,000}$ = 3.038

Considering the data provided in questions, the PVAF are at a discount rate of 12%.

∴Cost of capital = 12%

(d) NPV of project S = PV of cash inflows – PV of cash outflows

= (60,000 × $PVAF_{(12\%,4)}$) – 1,17,300 = (60,000 × 3.038) – 1,17,300

= ₹ 10,963.20

22. Modified Internal Rate of Return (MIRR)

It is based on compounding technique.

It assumes reinvestment of intermediate cash flows at cost of capital only.

Step – 1) Calculate total compound value of intermediate cash flows at end of project

Step – 2) Initial outflow × $(1 + r)^n$ = Total compound value

From above equation find r which is equal to MIRR

Or **MIRR** = $\sqrt[n]{\frac{Total\ Compound\ Value}{Initial\ outflow}} - 1$

Decision Criteria	Accept	Reject
MIRR	MIRR > COC	MIRR < COC

Question – 11

Calculate MIRR from the following data, if cost of capital is 9%:

Year	Cash Flows (₹)
0	1,50,000
1	40,000
2	70,000
3	90,000
4	30,000
5	50,000

Solution

Statement of Compound Value

Year	Cash Flows (₹)	CVF @ 9%	CV
1	40,000	1.412	56,480
2	70,000	1.295	90,650
3	90,000	1.188	1,06,920
4	30,000	1.090	32,700
5	50,000	1.000	50,000
		Total	3,36,750

MIRR = $\sqrt[5]{\frac{3,36,750}{1,50,000}} - 1 = 17.56\%$

23. Replacement Decisions

(a) Calculate Initial cash outflows

Particulars	Amount
Cost of new assets	-
(-) Sale value of old assets	-
(-) Tax saving on loss on sale of old assets	-
(+) Tax payment on profit on sale of old assets	-
(+) Increase in working capital	-
(-) Decrease in working capital	-
Cash Outflows	-

(b) Calculate Incremental Revenue CFs

Particulars	Amount
Increase in sales	-
(+) Savings in costs	-
(-) Increase in costs	-
Incremental PBD	-
(-) Increase in Depreciation (New – Old)	-
Incremental PBT	-
(-) Tax	-
Incremental PAT	-
(+) Incremental Depreciation	-
Incremental CFs	-

(c) Calculate incremental sale of assets at end and working capital realization

(d) Calculate NPV or IRR and take decision

Question – 12

Shiv Limited is thinking of replacing its existing machine by a new machine which would cost ₹ 60 lakhs. The company's current production is 80,000 units, and is expected to increase to 1,00,000 units, if the new machine is bought. The selling price of the product would remain unchanged at ₹ 200 per unit. The following is the cost of producing one unit of product using both the existing and new machine:

	Existing Machine (80,000 units)	New Machine (1,00,000 units)	Unit Cost (₹) Difference
Materials	75.0	63.75	(11.25)
Wages & Salaries	51.25	37.5	(13.75)
Supervision	20.0	25.0	5.0
Repairs & Maintenance	11.25	7.5	(3.75)
Power & Fuel	15.5	14.25	(1.25)
Depreciation	0.25	5.0	4.75
Allocated Corporate Overheads	10.0	12.5	2.5
	183.25	165.5	(17.75)

The existing machine has an account book value of ₹ 1,00,000, and it has been fully depreciated for tax purpose. It is estimated that machine will be useful for 5 years. The supplier of the new machine has offered to accept the old machine for ₹ 2,50,000. However, the market price of old machine today is ₹ 1,50,000 and it is expected to be ₹ 35,000 after 5 years. The new machine has a life of 5 years and a salvage value of ₹ 2,50,000 at the end of its economic life. Assume corporate income tax rate of 40% and depreciation is charged on straight line basis for income tax purposes. Further assume that book profit is treated as ordinary income for tax purpose. The opportunity cost of capital of the company is 15%. Required:

(a) Estimate net present value of the replacement decision

(b) Estimate the internal rate of return of the replacement decision

(c) Should company go ahead with the replacement decision? Suggest.

Year	1	2	3	4	5
$PVIF_{0.15,t}$	0.8696	0.7561	0.6575	0.5718	0.4972
$PVIF_{0.20,t}$	0.8333	0.6944	0.5787	0.4823	0.4019
$PVIF_{0.25,t}$	0.8000	0.6400	0.5120	0.4096	0.3277
$PVIF_{0.30,t}$	0.7692	0.5917	0.4552	0.3501	0.2693
$PVIF_{0.35,t}$	0.7407	0.5487	0.4064	0.3011	0.2230

Solution

(a) Statement of NPV

Particulars	Time	PVF	Amount	Present Value
Cost of new machine	0	1	60,00,000	60,00,000
(-) Cash flow from sale of old assets	0	1	(1,50,000)	(1,50,000)
			PVCO	**58,50,000**
Incremental Cash flows (w.n.-1)	1-5	3.3522	22,84,000	76,56,425
Incremental cash flow from sale of asset	5	0.4972	2,29,000	1,13,859
			PVCI	**77,70,284**
NPV (PVCI – PVCO)				**19,20,284**

Working Note – 1: Calculation of cash flow from sale of old assets

Book value of assets	-
Less: Sale value of assets (A)	2,50,000
Profit on sale	2,50,000
Tax @ 40% (B)	1,00,000
Cash from sale of old machine (A – B)	1,50,000

Working Note – 2: Calculation of cash flow from operations

Increase in sales [(1,00,000 – 80,000) × 200]	40,00,000
Less: Increase in cost	(9,60,000)
[{(63.75 + 37.5 + 25 + 7.5 + 14.25)×1,00,000} + {(75 + 51.25 + 20 + 11.25 + 15.5)×80,000}]	
Incremental PBD (A)	30,40,000
Less: Incremental Depreciation $\left[\left(\frac{60,00,000-2,50,000}{5}\right)-0\right]$	11,50,000
Incremental PBT	18,90,000
Tax @ 40% (B)	7,56,000
Incremental cash flow from operations (A – B)	22,84,000

Working Note – 3: Incremental cash flow from sale of assets

	New	Existing
Cost of assets	60,00,000	-
Accumulated Depreciation	57,50,000	-
Book Value	2,50,000	-
Sale Value (A)	2,50,000	35,000
Profit	-	35,000
Tax @ 40% (B)	-	14,000
Cash flow from sale of assets (A – B)	2,50,000	21,000

Incremental cash flow from sale of assets = 2,50,000 – 21,000 = ₹ 2,29,000

(b) **Statement of NPV**

Particulars	Time	Amount	PVF@20%	PV	PVF@30%	PV
Cost of new machine	0	60,00,000	1	60,00,000	1	60,00,000
(-) CF from old assets	0	(1,50,000)	1	(1,50,000)	1	(1,50,000)
		PVCO		**58,50,000**		**58,50,000**
Incremental CF	1-5	22,84,000	2.9906	68,30,530	2.4355	55,62,682
Incremental CF from asset	5	2,29,000	0.4019	92,035	0.2693	61,670
		PVCI		**69,22,565**		**56,24,352**
NPV (PVCI – PVCO)				**10,72,565**		**(2,25,648)**

IRR = 20% + $\left[\frac{10{,}72{,}565}{10{,}72{,}565-(-2{,}25{,}648)}\right](30-20)$ = 28.26%

(c) The company should go ahead with replacement project since it has positive NPV.

Question – 13

A company wants to invest in a machinery that would cost ₹ 50,000 at the beginning of year 1. It is estimated that the net cash inflows from operation will be ₹ 18,000 per annum for 3 years, if the company opts to service a part of the machine at the end of year 1 at ₹ 10,000 and the scrap value at the end of year 3 will be ₹ 12,500. However, of the company decides not to services the part, it will have to be replaced at the end of year 2 at ₹ 15,400. But in this case, the machine will work for the 4th year also and get operational cash inflow of ₹ 18,000 for the 4th year. It will have to be scrapped at the end of the year 4 at ₹ 9,000. Assuming cost of capital at 10% and ignoring taxes, will you recommend the purchase of this machine based on the net present value of its cash flows? If the supplier gives a discount of ₹ 5,000 for purchase, what would be your decision? (The present value factors at the end of years 0, 1, 2, 3, 4, 5 and 6 are respectively 1, 0.9091, 0.8264, 0.7513, 0.6830, 0.6290 and 0.5644).

Solution

Statement showing evaluation of mutually exclusive proposals

Particulars	**Time**	**P. V. Factor**	**Service Part**		**Replace Part**	
Cash Outflows:			**Amount**	**P. V.**	**Amount**	**P. V.**
Cost of Machinery	0	1	50,000	50,000	50,000	50,000
Service Cost	1	0.9091	10,000	9,091	----	----
(+) Replace Part	2	0.8264	----	----	15,400	12,727
P. V. of Cash Outflow (A)				59,091		62,727
Cash Inflows						
Cash Inflow from Operation	1-3	2.4869	18,000	44,764	---	---
	1-4	3.1699	----	-----	18,000	57,058
	3	0.7513	12,500	9,391	----	----
Scrap Value of Machine	4	0.6830	-----	------	9,000	6,147
P. V. of Cash Inflows (B)				54,155		63,205
NPV [(B) – (A)]				(4,936)		478

Advise:- Purchase machine & Replace the part at end of second year.

(ii) If the supplier gives a discount of ₹ 5,000 on purchase of machine

Proposals	**Service Part**	**Replace Part**
NVP	64	5,478
Cumulative	2.4869	3,1699
Equivalent Annual NPV	25.73	1,728

Advise: Purchase machine & Replace the part at end of second year.

Question – 14

An existing company has a machine which has been in operation for two years, its estimated remaining useful life is 4 years with no residual value in the end. Its current market value is ₹ 3 lakhs. The management is considering a proposal to purchase an improved model of a machine gives increase output. The details are as under:

Particulars	**Existing Machine**	**New Machine**
Purchase Price	₹ 6,00,000	₹ 10,00,000
Estimated Life	6 years	4 years
Residual Value	0	0
Annual Operating days	300	300
Operating hours per day	6	6
Selling price per unit	₹ 10	₹ 10
Material cost per unit	₹ 2	₹ 2
Output per hour in units	20	40

Labour cost per hour	₹ 20	₹ 30
Fixed overhead per annum excluding depreciation	₹ 1,00,000	₹ 60,000
Working Capital	₹ 1,00,000	₹ 2,00,000
Income-tax rate	30%	30%

Assuming that - cost of capital is 10% and the company uses written down value of depreciation @ 20% and it has several machines in 20% block.

Advice the management on the Replacement of Machine as per the NPV method. The discounting factors table given below:

Discounting Factors	Year 1	Year 2	Year 3	Year 4
10%	0.909	0.826	0.751	0.683

Solution

Statement of NPV

Particulars	Time	PVF	Amount	Present Value
Cost of new machine	0	1	10,00,000	10,00,000
(+) Add. working cap. (2,00,000 – 1,00,000)	0	1	1,00,000	1,00,000
(-) Cash flow from sale of old assets	0	1	(3,00,000)	(3,00,000)
			PVCO	**8,00,000**
Incremental Cash flows (w.n.-1)	1	0.909	2,59,000	2,35,431
	2	0.826	2,50,600	2,06,996
	3	0.751	2,43,880	1,83,154
	4	0.683	2,38,504	1,62,898
Incremental working capital realization	4	0.683	1,00,000	68,300
			PVCI	**8,56,779**
NPV (PVCI – PVCO)				**56,779**

Since the incremental NPV is positive, thus existing machine should be replaced.

Working Note – 1: Calculation of profit before depreciation (PBD)

Particulars	Existing Machine	New Machine
Annual output	300 × 6 × 20 = 36,000	300 × 6 × 40 = 72,000
Sales @ ₹ 10 per unit	3,60,000	7,20,000
Less: Cost of operation		
Material @₹ 2 per unit	72,000	1,44,000
Labour	1800 × 20 = 36,000	1800 × 30 = 54,000
Fixed OHs	1,00,000	60,000
Profit before Depreciation	1,52,000	4,62,000

Thus, Annual Incremental Profit Before Depreciation = 4,62,000 – 1,52,000 = ₹ 3,10,000

Working Note – 2: Calculation of basis of depreciation

Particulars	Existing	After Replacement
Purchase price of existing	6,00,000	6,00,000
Less: Depreciation of Yr. 1	1,20,000	1,20,000
Less: Depreciation of Yr. 2	96,000	96,000
WDV of existing machine	3,84,000	3,84,000
Add: Purchase of new	-	10,00,000
Less: Sale of existing	-	3,00,000
Basis for Depreciation	3,84,000	10,84,000

Base for Incremental Depreciation = 10,84,000 – 3,84,000 = ₹ 7,00,000

Working Note – 3: Incremental cash flow from sale of assets

Particulars	Year 1	Year 2	Year 3	Year 4
Incremental PBD (A)	3,10,000	3,10,000	3,10,000	3,10,000
Incremental Depreciation (on Base of ₹ 7,00,000) (B)	1,40,000	1,12,000	89,600	71,680
Incremental PBT (A – B)	1,70,000	1,98,000	2,20,400	2,38,320
Tax @ 30% (C)	51,000	59,400	66,120	71,496
Incremental CFs (A – C)	2,59,000	2,50,600	2,43,880	2,38,504